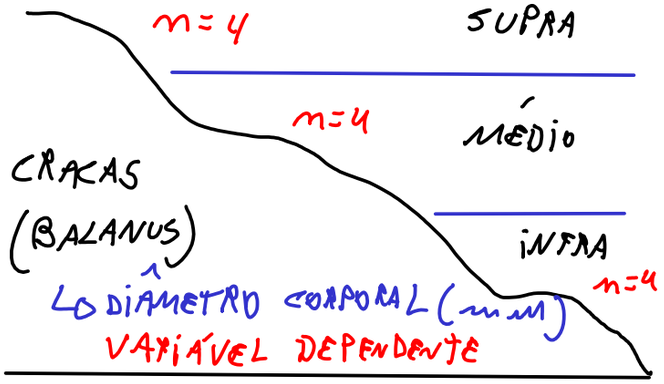
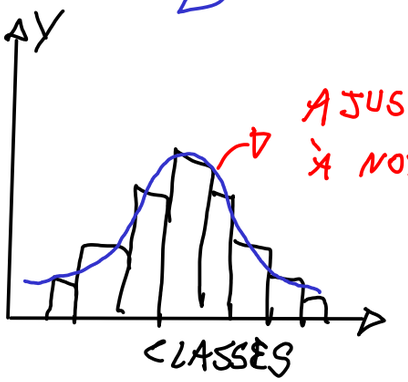
DIAM Y	ZONAS	SEXO
7	SUP	M
9	SUP	M
12	SUP	M
10	SUP	M
12	MED	F
9	MED	F
7	MED	M
11	MED	F
10	INF	F
9	INF	M
7	INF	F
12	INF	F

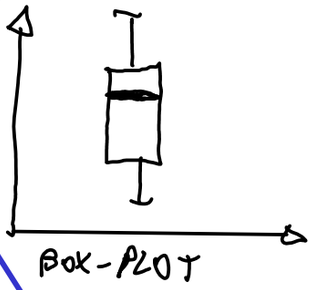
VARIÁVEL DEPENDENTE
OU VAR. RESPOSTA
VARIÁVEL ALEATÓRIA
OU Y

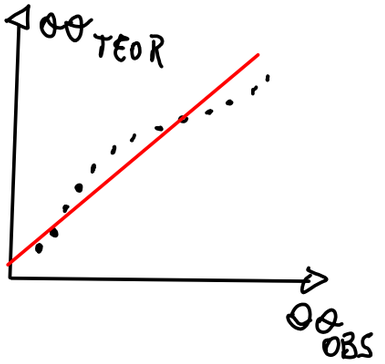


EXPLORAR



MÉTODOS GRÁFICO





TESTE DE HIPÓTESES

SHAPIRO $\alpha = 0,05$

$$\begin{cases} H_0: \text{DIST}_{OBS} = \text{DIST}_{NORMAL} \\ H_1: \text{DIST}_{OBS} \neq \text{DIST}_{NORMAL} \end{cases}$$

$P > 0,05$ NORMAL

$P < 0,05$ Ñ-NORMAL

TRANSFORMAÇÃO ($\log$, $\sqrt{\ }$, $\sqrt[3]{\ }$)

TESTES Ñ-PARAMÉTRICOS (LIVRES DE DISTRIBUIÇÃO)

DADOS NÃO-NORMAIS:

POSSÍVEL CAUSA: n pequeno.

TEOREMA DO LIMITE CENTRAL

$\uparrow n$ + NORMAL

BASEADO EM POSTOS (RANKS)

EXEMPLO:

VAR. ORIGINAL:

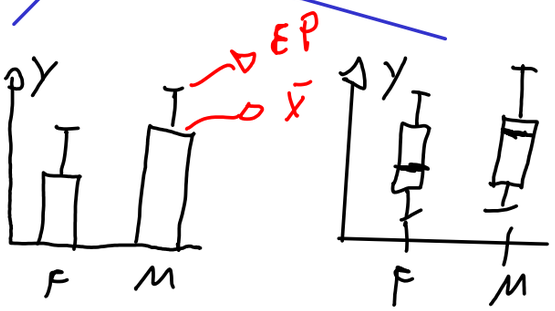
1 5 18 90 150

POSTOS

1 2 3 4 5

2 GRUPOS (CATEGORIAS) OU VAR. INDEPENDENTE OU FATOR (2 NÍVEIS)

EXPLORAR



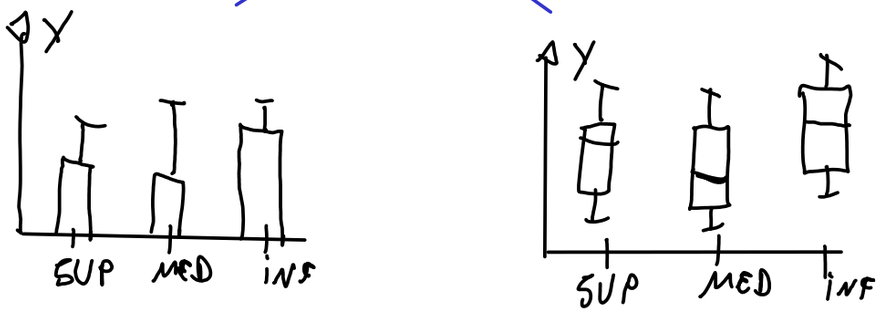
TESTE DE HOMOGENEIDADE DE VARIAÇÃO (HOMOCEDASTICIDADE)

BARTLETT

$$\begin{cases} H_0: \sigma_M = \sigma_F \\ H_1: \sigma_M \neq \sigma_F \end{cases}$$

3 GRUPOS OU +

EXPLORAR



TESTE DE HOMOGENEIDADE DE VARIAÇÃO (HOMOCEDASTICIDADE)

BARTLETT

$$\begin{cases} H_0: \sigma_S = \sigma_M = \sigma_I \\ H_1: \text{PELO MENOS } 1\ \sigma \text{ É } \neq \end{cases}$$

$P < 0{,}05$

HETEROC.

TESTE DE 2 MÉDIAS.

$\boxed{\text{TESTE-Z HETEROC.}}$

$P > 0{,}05$

HOMOCED.

TESTE DE 2 MÉDIAS

$\boxed{\text{TESTE-Z HOMOCED.}}$

$$Z = \frac{\bar{X}_M - \bar{X}_F}{EP}$$

TESTE BICAUDAL

$$\begin{cases} H_0: \mu_M = \mu_F \\ H_1: \mu_M \neq \mu_F \end{cases}$$

MONO-CAUDAL

ou $H_1: \mu_M > \mu_F$

ou $H_1: \mu_M < \mu_F$

$P < 0{,}05$

$\boxed{M \neq F}$

$P > 0{,}05$

$\boxed{M = F}$

$P < 0{,}05$

HETEROC.

TRANSFORMAÇÃO

$(\text{LOG}, \sqrt{\ }, \sqrt{\sqrt{\ }})$

$P > 0{,}05$

HOMOCED.

RÉPLICAS INDEPENDENTES

TESTE DE 3 MÉDIAS OU +

$\boxed{\text{ANOVA}}$

$$\begin{cases} H_0: \mu_S = \mu_M = \mu_I \\ H_1: \text{pelo menos 1 é} \neq \end{cases}$$

$P < 0{,}05$

REJEITO H_0

POST-HOC

OU A POSTERIORI

$\boxed{\text{TESTE DE TUKEY}}$

$P_{S \times M} =$

$P_{S \times I} =$

$P_{M \times I} =$

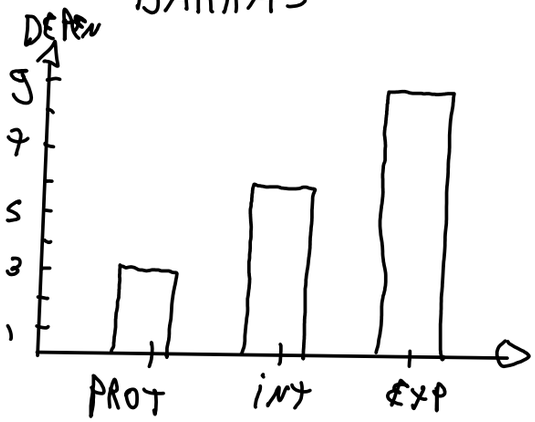
$P > 0{,}05$

ACEITO H_0.

EXERCÍCIO DE ANOVA

DEPEN	PRAIA
3	PROT
5	PROT
2	PROT
2	PROT
5	INT
7	INT
6	INT
6	INT
5	EXP
12	EXP
7	EXP
12	EXP

GRÁFICO DE BARRAS



TABELAMENTO

TRATAMENTO

RÉPLICAS

	PROT	INT	EXP
1	3	5	5
2	5	7	12
3	2	6	7
4	2	6	12
$\bar{X}$	3	6	9

GRANDE MÉDIA 6

$$VAR_{TOT} = VAR_{\substack{ENTRE\\TRAT}} + VAR_{\substack{DENTRO\\TRAT\\(RESÍDUO)}}$$

$$\text{ENTRE TRAT} = \left[\underbrace{(3-6)^2}_{9} + \underbrace{(6-6)^2}_{0} + \underbrace{(9-6)^2}_{9}\right] \times 4 = 18 \times 4 = \boxed{72}$$

$$\text{DENTRO TRAT (RESÍDUO)} = \underbrace{(3-3)^2}_{0} + \underbrace{(5-3)^2}_{4} + \underbrace{(2-3)^2}_{1} + \underbrace{(2-3)^2}_{1} + = 6$$

$$\underbrace{(5-6)^2}_{1} + \underbrace{(7-6)^2}_{1} + \underbrace{(6-6)^2}_{0} + \underbrace{(6-6)^2}_{0} + + = 2$$

$$\underbrace{(5-9)^2}_{16} + \underbrace{(12-9)^2}_{9} + \underbrace{(7-9)^2}_{4} + \underbrace{(12-9)^2}_{9} = + = 38$$

$$\boxed{46}$$

TABELA DE ANOVA

	S.Q.	G.L.	M.Q.	F	P
ENTRE TRAT	72	3-1=2	36	36/5,1 = 7	P<0,05
RESÍDUO	46	12-3=9	5,1		

POST-HOC TUKEY